Taxonomy and Structural Mechanisms of Hallucination in NLG (Ji et al. 2023)

Hallucination in neural language generation stems from parametric memory corruption, training data bias, and decoding divergence, categorizable into intrinsic and extrinsic factual inaccuracies.

GRADE: HIGH | STATUS: STABLE | DOMAIN: TECHNICAL, ORGANIZATIONAL, PHILOSOPHICAL | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Hallucination in natural language generation is not an anomalous bug but a structural feature of maximum-likelihood statistical sequence predictors. Models generate fluent, grammatically flawless statements that contradict the source context (intrinsic hallucination) or introduce unverifiable assertions absent from the training distribution or grounding input (extrinsic hallucination).

EVIDENCE AND CAUSAL ASSESSMENT

Causal roots of hallucination include: (1) imperfect memorization during pretraining, where co-occurrence statistics override factual precision; (2) exposure bias during autoregressive decoding; and (3) divergence between training data distribution and prompt requirements.

EFFECT SIZE AND PERSISTENCE

Hallucination rates range from 3% to over 30% depending on domain complexity, entity rarity, and context window length, persisting even as model scale expands into hundreds of billions of parameters.

BIAS, HETEROGENEITY, AND CONFOUNDING

Classified as: robust (<20%).

Classified as robust (<20%). Surface plausibility correlates inversely with detectability: larger models produce hallucinations that are more syntactically polished and self-consistent, increasing the cognitive load required to audit them.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Hallucination does not imply malicious intent or conscious deceit; it reflects probabilistic pattern completion operating across ungrounded parametric weight matrices.

WHAT ACTION IS REASONABLY JUSTIFIED

Never deploy ungrounded LLMs as authoritative single points of failure in regulated or mission-critical workflows; mandate external retrieval validation (RAG) and deterministic citation checks.

VERIFIED SOURCES & DOI IDENTIFIERS

- Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., Ishii, E., Bang, Y. J., Madotto, A., & Fung, P. (2023). Survey of Hallucination in Natural Language Generation. ACM Computing Surveys, 55(12), 1–38. https://doi.org/10.1145/3571730